

Corporate Finance

Article

Calculating ROI for technology investments

International Masters

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ROI for technology investments

- Value case or Business Case Analysis (BCA)
- Can be also called “Return on Investment” analysis...
- Based on a Cost / benefits analysis (CBA)
- Goal is to estimate the **financial value of a project**
- Used in the Go / NoGo decision for the project and / or in prioritization process...
 - ...To ensure company resources are spent on the most “profitable” investments
 - ...To avoid subjective selection / prioritization and managers “pet projects”

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- Objections to BCA :

- Hard to do... Requires large effort...
 - It requires “some” effort not necessarily large....
 - Does not seem reasonable to spend company money without any idea of the investment return
 - Even a high level BCA is better than no BCA at all...
 - It is an educational process part of good project governance....
- Subject to “gaming”....
 - Good governance reduces the risk of gaming...
 - Example : process driven by independent party (project office), clear and transparent rules applying to all projects, review of business cases by key stakeholders and management....

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- Producing BCA :
 - Assess the future project benefits (ex : productivity savings, revenue generation, cost savings....)
 - Assess project costs
- 4 Steps =
 - 1. Forecast benefits : qualify then quantify....
 - 2. Estimate project costs
 - 3. Map benefits and cost projections
 - 4. Calculate key financial metrics
 - Net benefits
 - ROI
 - Payback period
 - Net Present value
 - IRR – Internal rate of return
 -

ROI for technology investments

- Total Net Benefit
 - Difference between (Sum of the Benefits) – (Sum of costs)

	Year 0	Year 1	Year 2	Year 3	Year 4
Benefits	B1	B2	B3	B4	B5
Costs	C1	C2	C3	C4	C5
Net Benefits	B1-C1	B2-C2	B3-C3	B4-C4	B5-C5

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- Qualify the benefits brought by the project...
- Example : Purchase of a design SW will produce 2 benefits :
 - 1. Decrease the time required to build product
 - 2. Allow the building of a new product

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- Quantify Benefit 1 : decrease the time required to build product
 - Compare :
 - Baseline or current state, ie what's happening now without the project
 - 5,000 hours required per year
 - Each hour cost 15\$
 - Future state with the project :
 - Hours expected to decrease to 4,500 in Year 1 then 3,800 the following year
 - Same cost

	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
Hours Required--Current State	0	5,000	5,000	5,000	5,000	5,000
Costs: Current State at \$15/hour	\$0	\$75,000	\$75,000	\$75,000	\$75,000	\$75,000
Hours Required--Future State (Hours X Rate)	0	4,500	3,800	3,800	3,800	3,800
Costs: Future State at \$15/hour	\$0	\$67,500	\$57,000	\$57,000	\$57,000	\$57,000
BENEFIT #1: Reduced Labor Costs*	\$0	\$7,500	\$18,000	\$18,000	\$18,000	\$18,000

* (Future State Costs - Current State Costs)

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- Quantify Benefit 2 : allow the building of a new product
 - Compare
 - Baseline or current state, ie what's happening now without the project
 - Nothing as product does not exist...
 - Future state with the project
 - Estimation that we'll sell 2,000 products in Year 1 then 2,500 Year 2, then 3,000 following years
 - Margin per product = 10\$ per unit...

	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
New Products Sold--Current State	0	0	0	0	0	0
Net Revenue, Current State	\$0	\$0	\$0	\$0	\$0	\$0
New Products Sold--Future State	0	2,000	2,500	3,000	3,000	3,000
Net Revenue, Future State at \$10/unit	\$0	\$20,000	\$25,000	\$30,000	\$30,000	\$30,000
BENEFIT #2: Increased Net Revenue*	\$0	\$20,000	\$25,000	\$30,000	\$30,000	\$30,000

*(Future State Increased Net Revenue - Current State Increased Net Revenue)

- Summary of benefits :

*(Future State Increased Net Revenue - Current State Increased Net Revenue)

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- Project costs :

	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
Software Acquisition	\$100,000	\$0	\$0	\$0	\$0	\$0
Software Maintenance	-	\$15,000	\$15,000	\$15,000	\$15,000	\$15,000
Total Software Cost	\$100,000	\$15,000	\$15,000	\$15,000	\$15,000	\$15,000
Additional Employee Hours Required	1,000	150	50	50	50	50
Additional Labor Cost at \$15/hour	\$15,000	\$2,250	\$750	\$750	\$750	\$750

ROI for technology investments

- Example : Purchase of a design SW will produce 2 benefits :
 - Cost / Benefit analysis

Benefits		Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
Reduced Labor Costs		\$0	\$7,500	\$18,000	\$18,000	\$18,000	\$18,000
Increased Gross Profit		\$0	\$20,000	\$25,000	\$30,000	\$30,000	\$30,000
Total Benefits		\$0	\$27,500	\$43,000	\$48,000	\$48,000	\$48,000

Costs		Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
Software + Maintenance		\$100,000	\$15,000	\$15,000	\$15,000	\$15,000	\$15,000
Labor		\$15,000	\$2,250	\$750	\$750	\$750	\$750
Total Costs		\$115,000	\$17,250	\$15,750	\$15,750	\$15,750	\$15,750

Annual Net Benefits		Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
Net Benefits (Benefits - Costs)		(\$115,000)	\$10,250	\$27,250	\$32,250	\$32,250	\$32,250

ROI for technology investments

Net Benefits		Year 0	Year 1	Year 2	Year 3	Year 4	Year 5	Total
Net Benefits (Benefits - Costs)		(\$115,000)	\$10,250	\$27,250	\$32,250	\$32,250	\$32,250	\$19,250

ROI for technology investments

- **Questions :**

- **What is the value of the total net benefits ?**
 - +19,250
- **Based on this would you recommend moving forward with the project ?**
 - Yes, as the total net benefits is positive
- **If we apply a Discounted cash flow methodology (NPV) , would you expect the NPV to be higher or lower than to total net benefit above ?**
 - It will be lower as Discounted cash flows are always lower than future cash flows

ROI for technology investments

- **Questions :**

- If we define Payback period by the number of years it takes to recover the initial investment, what will be the Payback period for this project ?
 - Between year 4 and year 5

Payback Period		Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
Net Benefits (Benefits - Costs)		(\$115,000)	\$10,250	\$27,250	\$32,250	\$32,250	\$32,250
Running Total		(\$115,000)	(\$104,750)	(\$77,500)	(\$45,250)	(\$13,000)	\$19,250

Payback occurs 

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- Different Financial metrics to assess BCA
 - **Return on investment** = $(\text{Benefits} - \text{Costs}) / \text{Costs}$
 - On our example = $19,250 / 195,250$ or 9,9%

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- NPV : Net present value
 - Present Value
 - $PV = FV / (1+i)^n$

Calculating NPV	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5	Total
Net Benefits (Benefits - Costs)	(\$115,000)	\$10,250	\$27,250	\$32,250	\$32,250	\$32,250	\$19,250
Discount Factor with Hurdle Rate=10%	1.00	1.10	1.21	1.33	1.46	1.61	
Discounted Value	(\$115,000)	\$9,318	\$22,521	\$24,230	\$22,027	\$20,025	(\$16,879)

In this case, the projected NPV or total net benefit reflecting the time value of money of the proposed investment is **minus \$16,879**, not \$19,250. A proposed project with an estimated $NPV < 0$ is by definition a poor investment.

ROI for technology investments

- **Payback period** : time it takes for the benefits to outweigh the costs

Payback Period		Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
Net Benefits (Benefits - Costs)		(\$115,000)	\$10,250	\$27,250	\$32,250	\$32,250	\$32,250
Running Total		(\$115,000)	(\$104,750)	(\$77,500)	(\$45,250)	(\$13,000)	\$19,250

Payback occurs 

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- **Profitability index** : Future cash flows (Discounted or not) / Initial investment
 - On the example with discounted cash flows : $98,121 / 115,000 = 85\%$ (<1 , not good investment)

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- **IRR : Internal Rate of Return**

- Hurdle rate that would result in $NPV = 0$
- Calculated through trial and error method...
...or on a spreadsheet or calculator
- The higher, the better...
- On the example $IRR = 4,76 \%$
- To find a similar rate of return, company would need to find an investment that returns 4,76%

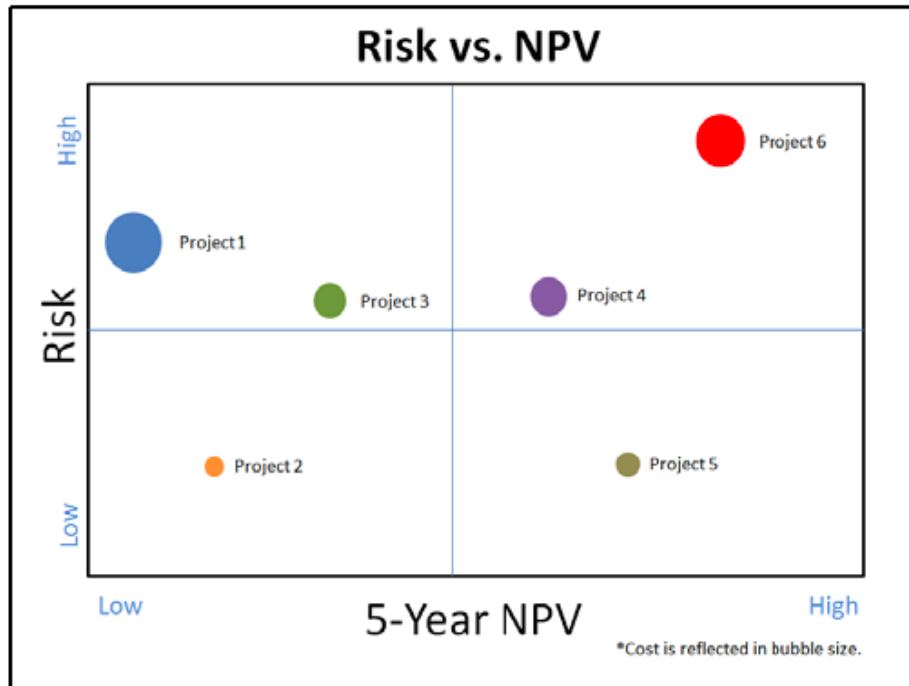
ROI for technology investments

- BCA is one of the elements to be taken into consideration :

Projects Under Consideration		Project A	Project B	Project C	Project D	Project E
	Net Benefits	\$775,000	\$245,000	\$450,000	\$2,275,000	\$573,000
	NPV	\$505,243	\$172,902	\$54,048	\$1,510,174	\$409,938
	ROI	76%	75%	20%	86%	85%
	Payback Period (Months)	19	11	44	26	9
	Profitability Index	488%	590%	145%	1010%	673%
	Upfront Costs	\$200,000	\$50,000	\$1,000,000	\$250,000	\$100,000
	Risk	Low	Low	High	Very High	Medium

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- NPV vs Risk



In this example, Project 5 has a relatively high NPV, relatively low projected costs, and one of the lowest measures of risk. For many companies, it is the best candidate for investment. A company with a strong risk appetite might consider Project 6, which has the highest projected NPV, albeit with the greatest risk and second highest projected costs. Project 1, with the lowest projected NPV, the second highest risk, and the highest projected costs, is the worst candidate for investment.

ROI for technology investments

- **BCA is an element in the decision making, other include cost and risk...**
 - Often a combination of Financial (ROI, NPV...), cost and risk is used...
- **Good practices :**
 - Be conservative in benefits and costs assessment
 - Be transparent on the assumptions used
 - Use various scenarios
 - Iterate...
 - Follow up with post implementation results
- **KISS – Keep it Simple Stupid ! and use common sense... don't get drowned in technicalities....**



Thank you!

See you next time